Low-p_T direct photon production in Run14 Cu+Au collisions at $\sqrt{s_{NN}} = 200$ GeV

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Abstract

Template note: this file was adapted from an older direct-photon analysis note. All figures, tables, numerical values, run lists, paths, captions, and conclusions must be replaced or revalidated for the Cu+Au analysis.

This analysis note describes methods for extracting low-p_T direct photon spectrum via external conversion method in Cu + Au system (Run12) at $\sqrt{s_{NN}} = 200$ GeV.

Contents

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1 Analysis Organization

Location of files and code:

- Directory for Cu + Au Taxi code:
  offline/AnalysisTrain/PCAl
- The path to run Cu + Au Taxi macro:
  offline/AnalysisTrain/pat/macro/Run_Run14CuAuPCA.C
- Directory for the Taxi output:
  /phenix/plhf/tongzhouguo/taxi/
- The path to analysis code and macros, which are used to analyze the Taxi output:
  /phenix/plhf/tongzhouguo/Pasha/dirphotCuAu
- The path to the final data files and macros to produce physics plots:
  /phenix/plhf/tongzhouguo/Pasha/dirphotCuAu

Brief summary of the analysis:
Study of low-p_T direct photons in Cu + Au system at 200 GeV

Data set and event selection:
Minimum bias Run-14 200 GeV Cu+Au pro102 ($\sim 2.9 \cdot 10^9$ events)

Analysis cuts:

1. Event selection: $|z_{\rm vtx}| < 20$ cm, trigger BBCLL1(> 0 tubes)_narrowvtx.
2. Track selection:
   - $quality = 31|51|63$
   - $|zed| < 75$ cm
   - DC deadmap
3. e^+/e^-:
   - $p_T > 0.3$ GeV
   - $n_0 \geq 1$
   - $disp < 5$
   - $\chi^2/npe0 < 10$
   - $E_{\rm core} > 0.4$ GeV
   - $prob > 0.01$
   - $E/p > 0.5$
   - $|emcdz| < 20$ cm and $|emcd\varphi| < 0.05$ rad

- EMCal deadmap

4. Conversion pair selection: dzed cut (see. 3.2.5)
5. EMCal photon clusters:
   - $E_{\mathrm{core}} > 0.4$ GeV
   - $\chi^2 < 5$
   - e^+/e^- veto (see. 3.2.6)
   - EMCal deadmap

1.1 Analysis Code Organization

Module PCAl is a fork from PhotonConversionAnalysis module. The output from the taxi is a ROOT TTree, filled with custom-class events that contain tracks, clusters and conversion pairs. By default we fill events only when there is at least one conversion pair reconstructed. Macros

```
trees/first_TTree.C
trees/second_TTree.C
trees/third_TTree.C
```

are then used to obtain foregrounds and backgrounds (via event-mixing) of $M_{e^+e^-}$ and $M_{e^+e^-\gamma}$ (see 3.3 and 3.4). Outputs are merged together and hists are produced with macro

```
dump_hists_simCent.C
```

Histograms are then used in macro

```
plot_histograms.C
```

to produce N_{incl} and N_{tag} and their ratio $N_{\mathrm{incl}}/N_{\mathrm{tag}}$.

Simulation chain for $\langle \varepsilon_\gamma f \rangle$ is as follows: generate π^0 via macro

```
sim/make_sim/make_single_pi0.C
```

and feed OSCAR output into PISA and then PISAToDST. Then use same chain as real data to get TTrees with PCAl module. Outputs are then used in macro

```
trees/embed_TTree.C
```

to calculate $M_{e^+e^-}$ and $M_{e^+e^-\gamma}$. Histograms are produced with

```
dump_hists_simCent.C
```

and are then used in macro

```
plot_histograms.C
```

to produce N_{incl} and N_{tag} and their ratio $N_{\mathrm{tag}}/N_{\mathrm{incl}}$.

Cocktail ratio γ^h/γ^{π^0}: EXODUS cocktail library with input π^0 spectrum. Input spectrum parameters are to be defined in

```
sim/exodus/cocktail_mt/initializeGenerators.h
```

and then make

```
sim/exodus/cocktail_mt/cocktail
```

Then macro

```
get_cocktail.C
```

is used to obtain cocktail ratio and γ^h spectrum.

Once all the ingredients are calculated, they are used together in macro

```
plot_rgamma.cpp
```

to get R_γ and direct photon spectrum and produce final plots.

2 QA

2.1 EMCal

2.1.1 Dead Maps

In this analysis we use only PbSc sectors of EMCal. The procedure of obtaining dead maps is described in [3]. Here we use same maps and same sector numbering scheme (see fig. 2.1.1).

2.1.2 Run-by-Run and Sector-by-Sector Energy Scale Calibration

Energy scale calibrations are done separately for each sector and each run. We first collect all of the clusters of EMCal in a given event, then select only those that pass following cuts:

- $E_{core} > 0.3$ GeV,
- $\chi^2 < 5$

Next we calculate invariant mass of every cluster pair combination with energy asymmetry $\alpha < 0.8$ and plot 2D histogram with mass axis and pair p_T axis (foreground). We then project $p_T > 3$ GeV part of this histogram onto mass axis.

Since our mass distribution will contain uncorrelated photon pairs, there will be background together with signals of π^0-meson. Event mixing technique is utilized to estimate the shape of this background: we pick two events with similar centralities and pair photon $\boldsymbol{\gamma}_1$ from event 1 with photon $\boldsymbol{\gamma}_2$ from event 2. We scale background to foreground in side band $[0.3; 0.4]$ GeV mass range to avoid η-meson signal, and subtract it from foreground.

Resulting distribution is then fitted with the sum of Gaussian function and 2nd order polynomial. Extracted Gaussian function parameters are used as mean mass of π^0 and its width. We calculate scale factor as ratio of 0.138 GeV$/c^2$ over extracted mass value.

See appendix A for comparison.

2.1.3 Bad Run Rejection

Calorimeter sectors may have different performance during different data acquisition periods. To ensure that we have adequate behavior across our whole sample, we need to filter out data segments where some performance issues are noticed. These data segments are marked as ”bad” and removed from subsequent analysis. We use same list of bad runs for EMCal as in [7].

2.2 DC

2.2.1 Dead Maps

We use DC dead maps from ϕ analysis [4] with no run dependence.

2.2.2 Beam Offset Correction

The inclination angle α shows how charged track "bends" in magnetic field before DC. When there is no magnetic field, charged particles should have $\alpha = 0$ independent of ϕ_{DC} (azimuthal angle ϕ of the track at DC reference radius $R_{\mathrm{DC}} = 220$ cm). We check this assumption by plotting $\langle\alpha\rangle$ as a function of ϕ_{DC} in available zero field runs (i.e. runs with no magnetic field). We see that $\langle\alpha\rangle$ is dependent on ϕ_{DC} for both east and west arms, which means that there is some misalignment of DC. To correct the alignment in physics runs we follow procedure described in [3] and fit the dependence with function

$$\alpha = \frac{dx}{R_{\mathrm{DC}}}\sin\phi_{\mathrm{DC}} + \frac{dy}{R_{\mathrm{DC}}}\cos\phi_{\mathrm{DC}} \tag{2.2.1}$$

and use extracted parameters dx and dy to calculate $\Delta\alpha$. See fig. 2.2.1 for comparison. Table 2.1 summarizes extracted parameters for zero field runs.

ZF Run Number	East Arm		West Arm	
	dx	dy	dx	dy
415835	-0.175	-0.036	0.173	0.050
416379	-0.180	-0.043	-0.015	0.039
416442	-0.186	-0.049	0.030	0.030

Table 2.1: Extracted $\alpha(\phi_{\mathrm{DC}})$ fit parameters for Cu + Au zero field runs **TODO: replace/check all numbers in this caption/table for Cu+Au.**

TODO: Check/replace the numeric values in the table above for Cu+Au.

2.2.3 Bad Run Rejection

Additionally to the list of EMCal bad runs we add list of DC bad runs. The procedure is as follows: pick a reference run (typically with largest statistics) and project DC occupancy maps onto board axis (and normalize). Next we divide each run projection by reference and fit with constant function. Large χ^2 fit value may indicate deviation of performance of DC in a given run. We then manually check outliers and reject runs listed in tab. 2.2

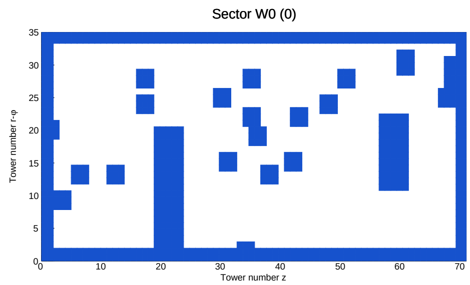
TODO: Check/replace the numeric values in the table above for Cu+Au.

Rejected Run Numbers	Description
415752, 415822, 415823, 415825, 415829, 415830	Dead areas in DCE not captured by dead maps
415906, 415927, 416018, 416361, 416361, 416367, 416374, 416429, 416435, 416436, 416540, 416620, 416627, 416628, 416632, 416688, 416695, 416810, 416888, 416891	Low relative hit rate in either DCE, DCW or both

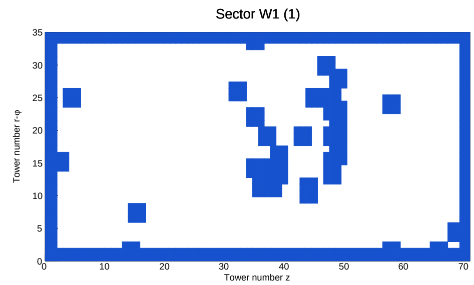
Table 2.2: Additional rejected runs **TODO: replace/check all numbers in this caption/table for Cu+Au.**

2.3 Momentum Scale Recalibration

We checked proton mass using TOF, and found that it is 3% higher than the PDG value, which means that reconstructed charged tracks will have higher momentum. To account for that we apply 0.97 scaling factor on momentum of all charged tracks. The details of procedure are described in [5].



TODO (Tongzhou): replace/check this figure for the Cu+Au direct-photon analysis.



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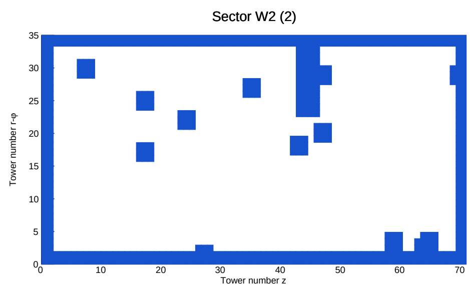

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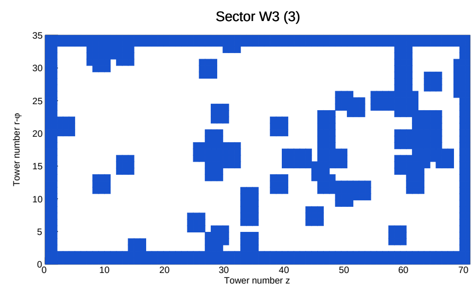

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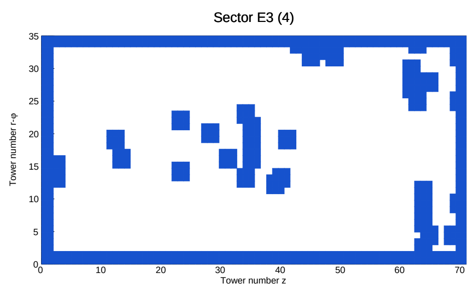

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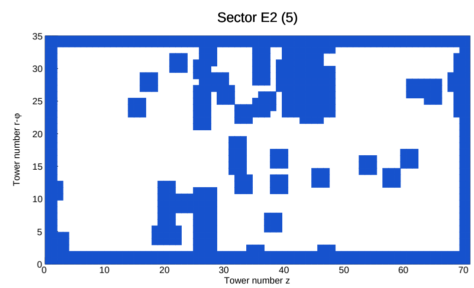
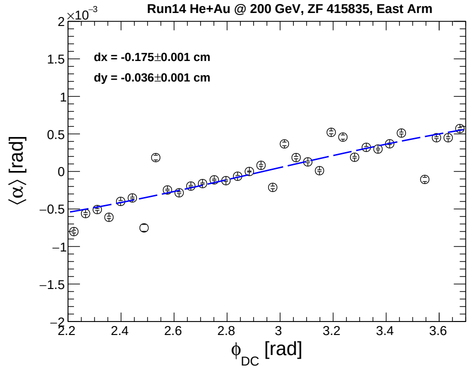
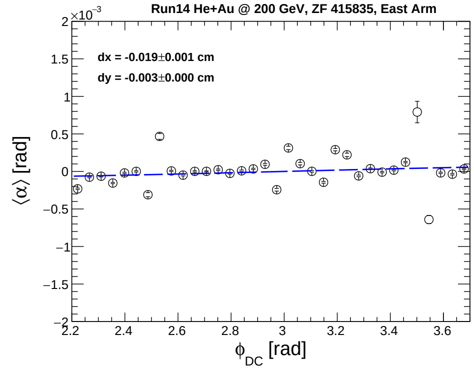


TODO (Tongzhou): replace/check this figure for the Cu+Au direct-photon analysis.

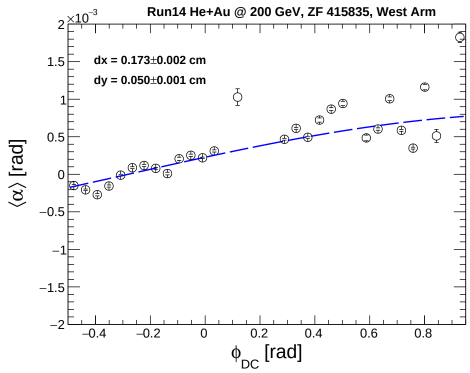
Figure 2.1.1: Dead maps for sectors 0-5 of EMCal, blue areas represent masked towers **TODO: replace/check all numbers in this caption/table for Cu+Au.**



TODO (Tongzhou): replace/check this figure for the Cu+Au direct-photon analysis.



TODO (Tongzhou): replace/check this figure for the Cu+Au direct-photon analysis.



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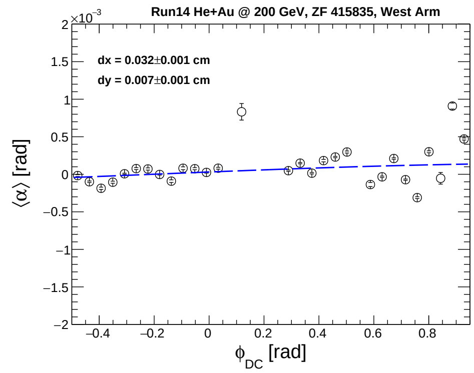


TODO (Tongzhou): replace/check this figure for the Cu+Au direct-photon analysis.

Figure 2.2.1: $\langle\alpha\rangle$ vs ϕ_{DC} for east and west arms in zero field run 415835. Left: before correction. Right: after correction **TODO: replace/check all numbers in this caption/table for Cu+Au.**

3 Data Analysis

3.1 External Conversion Method

As we are interested in low-p_T region, using only EMCal for photon analysis is insufficient due to its worsening resolution with decreasing energy. External conversion technique consists of reconstructing e^+e^- pairs from real photon conversions $\gamma \to e^+e^-$ in some material. Reconstructed momenta can be then used to obtain conversion photon momentum. Here we employ conversion reconstruction algorithm developed in [8].

3.1.1 Double Ratio

We approximate excess photon ratio R_γ with the following form referred as *double ratio*:

$$R_\gamma = \frac{\gamma^{\text{inclusive}}}{\gamma^{\text{decay}}} \approx \frac{\langle \varepsilon_\gamma f \rangle \left(\dfrac{N_{\text{incl}}}{N_{\text{tag}}} \right)_{\text{data}}}{\left(\dfrac{\gamma^h}{\gamma^{\pi^0}} \right)_{\text{MC}}} \tag{3.1.1}$$

The numerator represents the measured ratio of inclusive photons over π^0-decay photons. In the denominator we have ratio of all decay photons over π^0-decay photons, which is calculated via Monte-Carlo simulation.

We use π^0-decay photons in double ratio as they are the most abundant source of decay photons, and can be extracted by counting π^0 signals in $\pi^0 \to \gamma\gamma$ channel with the following procedure:

1. Extract inclusive photon sample (as real conversion pairs)
2. Combine conversion pairs with EMCal photons and calculate invariant mass distribution $M_{e^+e^-\gamma}$
3. Extract π^0 signal from $M_{e^+e^-\gamma}$
4. Since one photon comes from conversion sample and one from EMCal, the number of π^0-decay photons in inclusive sample is equal to the number of π^0 signals reconstructed from $M_{e^+e^-\gamma}$

This tagging procedure has efficiency (labeled as $\langle \varepsilon_\gamma f \rangle$), as we are not guaranteed to reconstruct every π^0 due to detector effects.

3.1.2 Invariant Yield

Once we have R_γ, calculating invariant yield of direct photons is straightforward: by definition we have $\gamma^{\text{inclusive}} = \gamma^{\text{decay}} + \gamma^{\text{direct}}$ and hence

$$\gamma^{\text{direct}} = \gamma^{\text{decay}}(R_\gamma - 1) \tag{3.1.2}$$

In this analysis we approximate γ^{decay} as $\gamma^{\text{decay}} \approx \gamma^h$ (see Sec. 3.6).

3.2 Cuts

3.2.1 Event Selection

The trigger we use is BBCLL1($>$ 0 tubes)_narrowvtx. We select events with $|z_{\text{vtx}}| < 20$ cm.

3.2.2 Charged Tracks

- $quality = 31|51|63$
- $|zed| < 75$ cm
- $p_T > 0.3$ GeV

3.2.3 Electron ID

To identify electrons out of all charged tracks we use RICH subsystem with the following cuts:

- $n_0 \geq 1$
- $disp < 5$
- $\chi^2/npe0 < 10$

Additionally we use EMCal variables:

- $E_{\text{core}} > 0.4$ GeV
- $E/p > 0.5$
- $prob > 0.01$
- $|emcdz| < 20$ cm and $|emcd\varphi| < 0.05$ rad.

3.2.4 EMCal Photon Clusters

For clusters we use $E_{\text{core}} > 0.4$ GeV and shower shape cut $\chi^2 < 5$.

3.2.5 Conversion Pair Cuts

We consider multiple conversion pair setups listed in tab. 3.1, we choose dzed cut as our central value.

Cut name	Description
loose cut	solution for reconstruction algorithm, e^- and e^+ in the same arm, $1 < r_{\text{conv}} < 29$ cm
dzed cut	loose cut, $\lvert dzed \rvert < 4$
default cut	dzed cut, $d\varphi_{\text{conv}} < 0.05$ mrad, $d\theta_{\text{conv}} < 0.02$ rad, $\phi_V < 0.5$
tight cut	loose cut, $\lvert dzed \rvert < 2$ cm, $9 < r_{\text{conv}} < 21$ cm, $d\varphi_{\text{conv}} < 0.05$ mrad, $d\theta_{\text{conv}} < 0.01$ rad, $\phi_V < 0.3$ rad
ϕ_V cut	loose cut, $\phi_V < 0.5$ rad

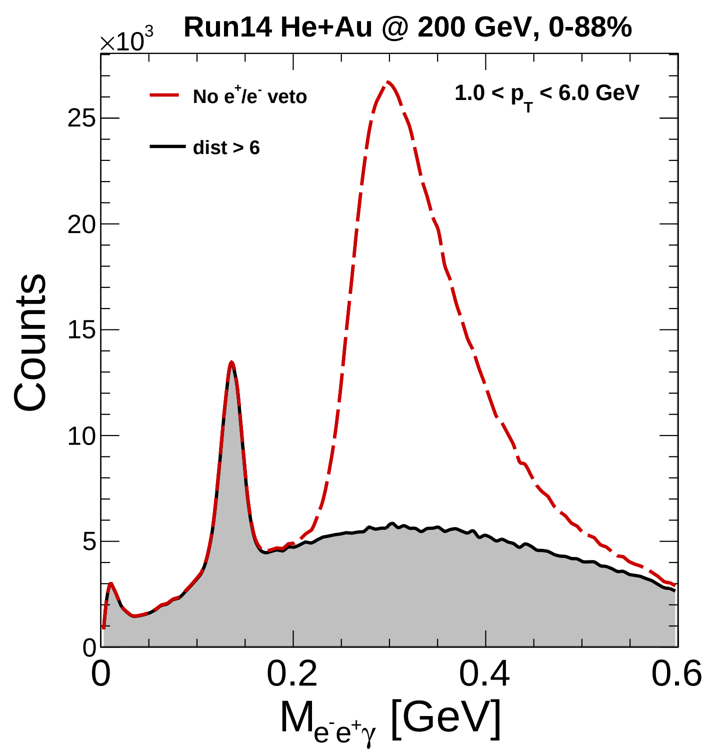
Table 3.1: Different conversion pair cut setups **TODO: replace/check all numbers in this caption/table for Cu+Au.**

TODO: Check/replace the numeric values in the table above for Cu+Au.

3.2.6 Electron/positron Veto

EMCal photon cluster cuts are insufficient for proper photon reconstruction, since the sample will contain clusters associated with conversion electrons and positrons. These clusters will result in correlated peak at around 0.3 GeV in $M_{e^+e^-\gamma}$ (see fig. 3.2.1), which we wish to avoid for better background scaling.

We define this veto by cutting on variable that quantifies the distance between the



TODO (Tongzhou): replace/check this figure for the Cu+Au direct-photon analysis.

Figure 3.2.1: Invariant mass distribution $M_{e^+e^-\gamma}$ plotted with e^+/e^- veto (black filled curve) and with no veto (red dashed line) **TODO: replace/check all numbers in this caption/table for Cu+Au.**

EMCal cluster and e^+/e^- track projection onto EMCal:

$$dist = \sqrt{\left(\frac{\Delta\varphi}{\sigma_{d\varphi}}\right)^2 + \left(\frac{\Delta z}{\sigma_{dz}}\right)^2} > 6, \tag{3.2.1}$$

where $\Delta\varphi$ is the difference of φ angles of track projection and cluster, dz — difference in z coordinates and σ_{dz} and $\sigma_{d\varphi}$ are taken from definitions of reduced variables $emcd\varphi$ and $emcdz$ (see appendix B).

3.3 Inclusive Sample

Inclusive sample consists of all real photons, that are reconstructed as e^+e^- conversion pairs. In this analysis we consider VTX detector material as the convertor. Since VTX is located at some radial distance from beam crossing point, we expect conversion tracks to originate from some radial point. This assumption is wrong in standard track reconstruction algorithm, for which all charged tracks originate directly from event vertex. The mismatch of origin point results in shifted momentum of conversion tracks. We can clearly see it by plotting invariant mass of conversion e^+e^- pairs (fig. 3.3.1). Without this momentum shifting, mass of the pair should equal to 0.

We calculate N_{inc} by integrating the distribution in range [0; 0.16] GeV after event mixing background subtraction. We use loose cut setup (see tab. 3.1) to scale event mixing background in range [0.2; 0.325] GeV, and use this ratio in other cut setups.

As we employ conversion reconstruction algorithm, we also correct conversions' momenta before mixing with EMCal photons.

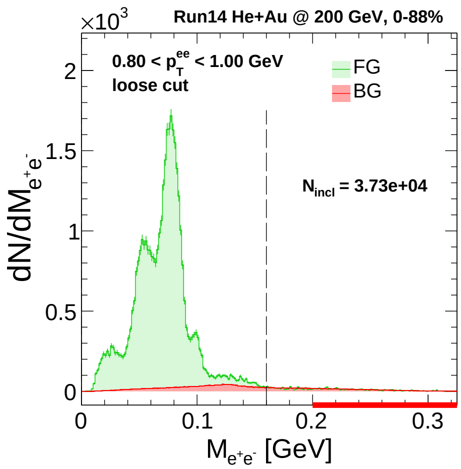
3.4 Tagged Sample

Tagged sample consists of π^0-decay photons that are "tagged" by reconstruction of π^0 signal from $M_{e^+e^-\gamma}$ distribution. We have one decay photon in our conversion sample and one in EMCal photon sample. This way one reconstructed π^0-meson corresponds to one decay photon in conversion sample.

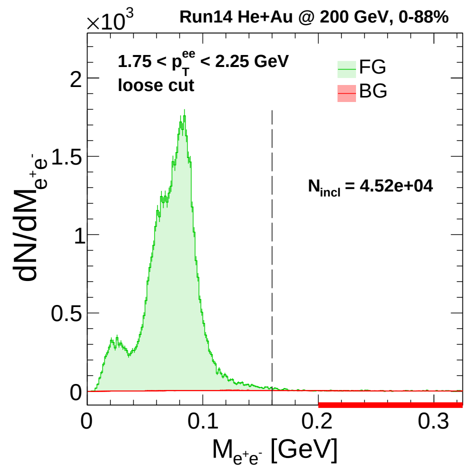
Since there is background in conversion sample, we can mix misreconstructed conversion photons ("fake conversions") with EMCal photons. There are two possible options (see [5, 6] for details):

1. One of the constituents (e^+/e^-) comes from true converted π^0-decay photon, yet it was paired with wrong partner to yield fake conversion. If the other π^0-decay photon is detected in EMCal, we expect to see some correlated part under π^0 peak when mixing these pairs (we denote it here as SG_{corr})

2. Fake conversion has no correlation with any of π^0-decay photons (denoted as BG_{corr})



TODO (Tongzhou): replace/check this figure for the Cu+Au direct-photon analysis.



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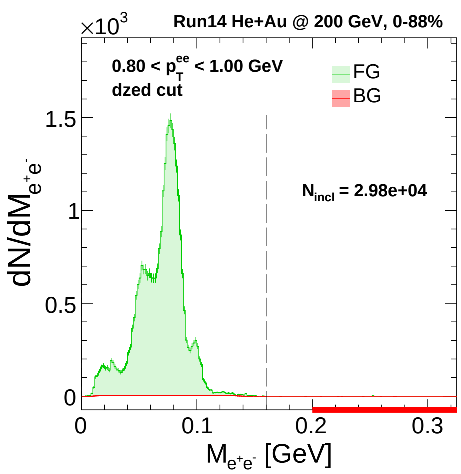

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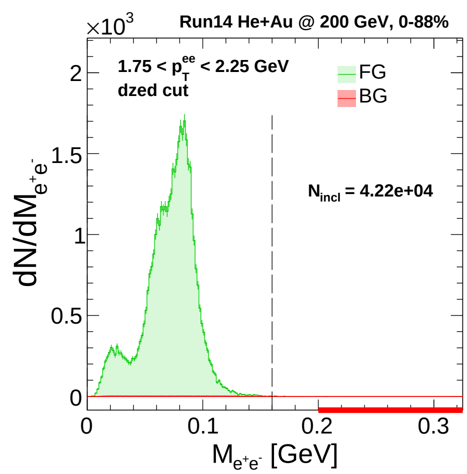
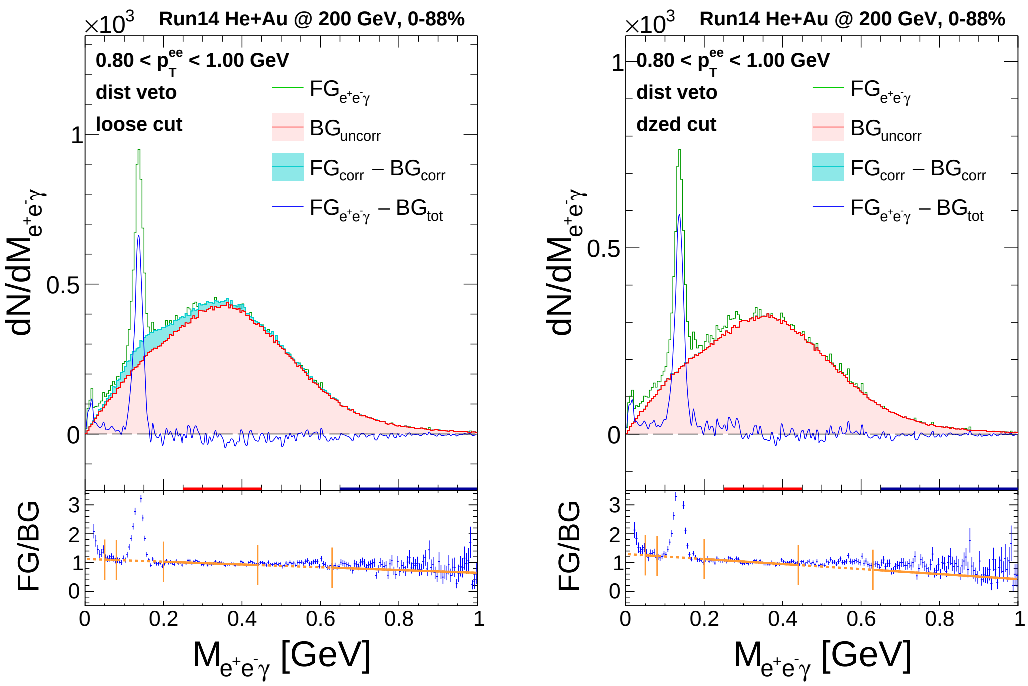


TODO (Tongzhou): replace/check this figure for the Cu+Au direct-photon analysis.

Figure 3.3.1: Invariant mass distributions $M_{e^+e^-}$ for two cut setups and two p_T ranges **TODO: replace/check all numbers in this caption/table for Cu+Au.**



TODO (Tongzhou): replace/check this figure for the Cu+Au direct-photon analysis.

TODO (Tongzhou): replace/check this figure for the Cu+Au direct-photon analysis.

Figure 3.4.1: Invariant mass distributions $M_{e^+e^-\gamma}$ for lowest p_T range for loose and dzed cuts **TODO: replace/check all numbers in this caption/table for Cu+Au.**

Correlated foreground $\mathrm{FG}_{\mathrm{corr}} = \mathrm{BG}_{\mathrm{corr}} + \mathrm{SG}_{\mathrm{corr}}$ is calculated as follows: pick an electron/positron from event 1 and combine it with positron/electron from event 2 to form "fake conversion", then mix this fake conversion with EMCal photons from event 2. The $\mathrm{BG}_{\mathrm{corr}}$ estimation is then straightforward: to remove any correlation we simply mix fake conversions with EMCal photons from event 3. To get the $\mathrm{SG}_{\mathrm{corr}}$ we scale $\mathrm{BG}_{\mathrm{corr}}$ to $\mathrm{FG}_{\mathrm{corr}}$ in side band [0.65; 1.00] GeV and subtract.

The amount of correlated signal $\mathrm{SG}_{\mathrm{corr}}$ is proportional to BG/FG ratio in inclusive sample, but since we mix fake conversions with EMCal photons only in one event (hence ignoring possible correlation with other pair constituent), we lose half of these correlated signals, therefore we additionally introduce a factor of 2 when scaling. Resulting $\mathrm{SG}_{\mathrm{corr}}$ is represented by cyan filled area on fig. 3.4.1.

Uncorrelated (combinatorial) background is estimated as follows: pick conversion pair from event 1 and mix it with EMCal photon from event 2. We scale uncorrelated background to $\mathrm{FG}_{\mathrm{total}}$ in side band [0.65; 1.00] GeV. Total foreground of $M_{e^+e^-\gamma}$ distribution is

$$\mathrm{FG}_{\mathrm{total}} = \mathrm{SG}_{\pi^0} + \mathrm{SG}_{\mathrm{corr}} + \mathrm{BG}_{\mathrm{uncorr}} \tag{3.4.1}$$

Once all background sources are subtracted we fit residual distribution with sum of Gaussian function and 2nd order polynomial. Then we calculate N_{tag} by counting π^0 signals in [0.09; 0.18] GeV window (after residual background subtraction).

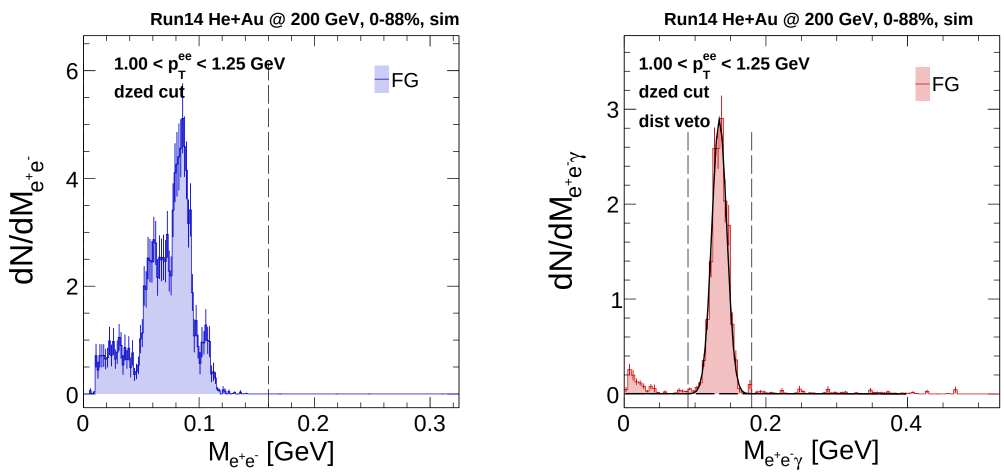
3.5 Tagging Efficiency

Tagging efficiency is labeled as $\langle \varepsilon_\gamma f \rangle$ in (3.1.1) and is defined as probability to tag second π^0-decay photon (by reconstructing π^0 with procedure described above), given that one of decay photons is already reconstructed as a conversion. There are indeed other efficiencies that are associated with conversion reconstruction, but they cancel in the ratio $(N_{\mathrm{incl}}/N_{\mathrm{tag}})$, so we are left with $\langle \varepsilon_\gamma f \rangle$.

Tagging efficiency can be calculated in Monte-Carlo setup by generating π^0-decay photon sample via

$$\langle \varepsilon_\gamma f \rangle = \left. \frac{N_{\mathrm{tag}}}{N_{\mathrm{incl}}} \right|_{\gamma^{\mathrm{inclusive}} = \gamma^{\pi^0}} \tag{3.5.1}$$

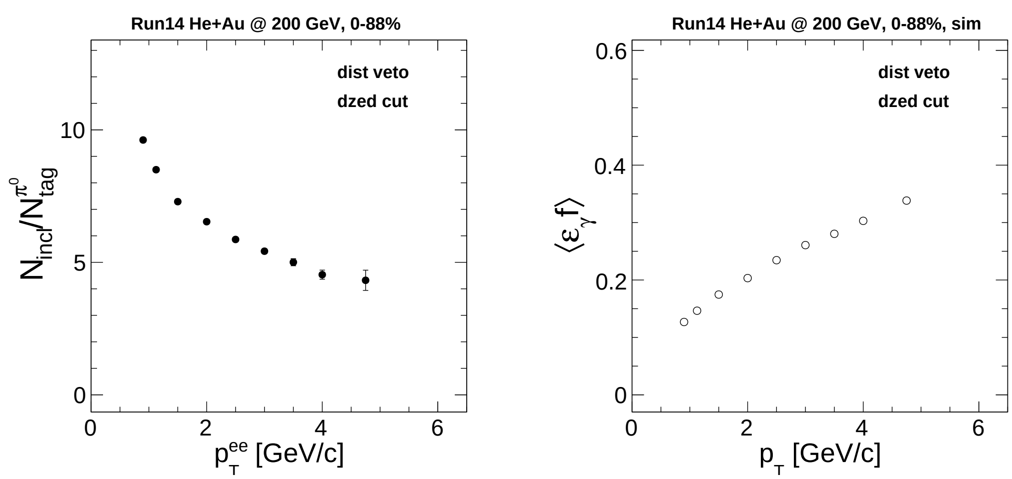
First we tune energy scale and resolution of EMCal to match data (see appendix D). Next generate spectrum of π^0 (flat in p_T and z_{vtx} distributed from real data). We then feed it into PISA and treat output as real data, while also having access to generated particle information. The resulting distributions are weighted by fitted modified Hagedorn π^0 spectrum (see appendix C).



TODO (Tongzhou): replace/check this figure for the Cu+Au direct-photon analysis.

TODO (Tongzhou): replace/check this figure for the Cu+Au direct-photon analysis.

Figure 3.5.1: Distributions $M_{e^+e^-}$ (left) and $M_{e^+e^-\gamma}$ (right) obtained from simulation **TODO: replace/check all numbers in this caption/table for Cu+Au.**



TODO (Tongzhou): replace/check this figure for the Cu+Au direct-photon analysis.

TODO (Tongzhou): replace/check this figure for the Cu+Au direct-photon analysis.

Figure 3.5.2: $N_{\mathrm{incl}}/N_{\mathrm{tag}}$ ratio (left) and tagging efficiency (right) for dzed cut **TODO: replace/check all numbers in this caption/table for Cu+Au.**

3.6 Cocktail Ratio

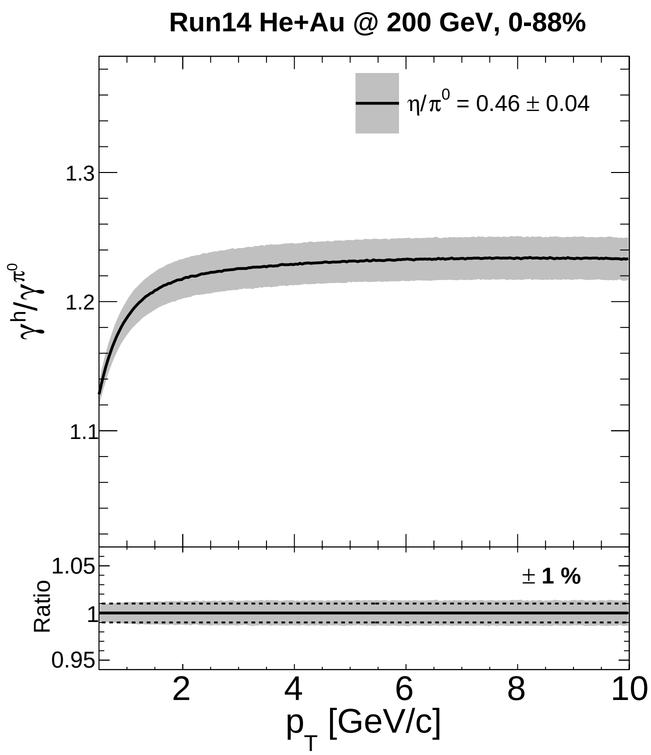
While there are too many decay photons sources to account for, we restrict ourselves to only a few that contribute the most. These include π^0, η, η' and ω mesons.

Hadron	Yield relative to π^0
η	0.46
η'	0.25
ω	0.90

Table 3.2: Particle yield relative to π^0 at 5 GeV **TODO: replace/check all numbers in this caption/table for Cu+Au.**

TODO: Check/replace the numeric values in the table above for Cu+Au.

We calculate the ratio using cocktail library built upon EXODUS, which has π^0 spectrum function as its input. Other particle spectra are obtained by m_T-scaling assumption and overall scale at 5 GeV, listed in tab. 3.2. We denote the total decay photon spectrum obtained from simulation as $\boldsymbol{\gamma}^h$, which is used both in denominator of (3.1.1) and for obtaining direct photon spectrum (3.1.2).



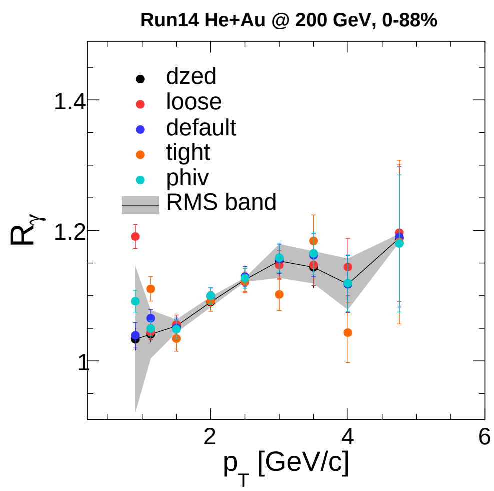
TODO (Tongzhou): replace/check this figure for the Cu+Au direct-photon analysis.

Figure 3.6.1: Cocktail ratio γ^h/γ^{π^0} **TODO: replace/check all numbers in this caption/table for Cu+Au.**

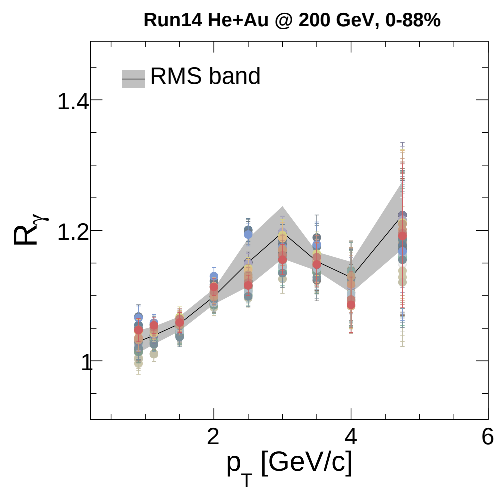
4 Systematic Uncertainties on R_γ

4.1 Conversion Sample Purity

To account for uncertainty due to different conversion cut selection criteria we calculate R_γ for all cut setups listed in 3.1. We use RMS of absolute deviations from central value (dzed cut) as systematic uncertainty (see fig. 4.1.1).



TODO (Tongzhou): replace/check this figure for the Cu+Au direct-photon analysis.



TODO (Tongzhou): replace/check this figure for the Cu+Au direct-photon analysis.

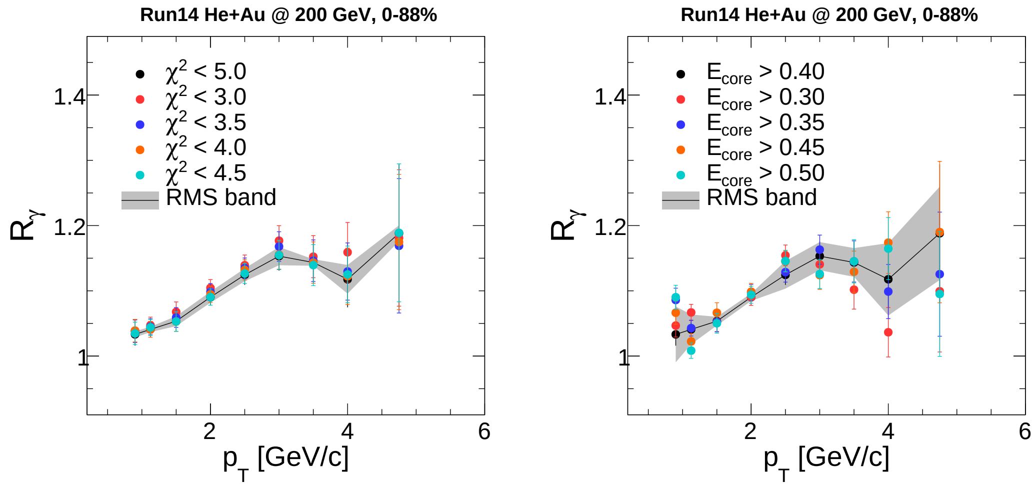
Figure 4.1.1: Variations of R_γ for different conversion pair cut setups (left) and π^0 peak extraction (right) **TODO: replace/check all numbers in this caption/table for Cu+Au.**

4.2 Energy Threshold

To see the effect of mismatch between energy scale and resolution in simulation and real data we vary E_{core} cut from 0.3 GeV to 0.5 GeV with 0.05 step (see fig. 4.3.1).

4.3 Photon Efficiency

The shower shape in EMCal might be different in simulation as compared to real data. To see the effect of this mismatch we vary χ^2 cut (see fig. 4.3.1).



TODO (Tongzhou): replace/check this figure for the Cu+Au direct-photon analysis.

TODO (Tongzhou): replace/check this figure for the Cu+Au direct-photon analysis.

Figure 4.3.1: Variations of R_γ for different shower shape cuts $\chi^2 < \alpha$ (left) and threshold cut on E_{core} (right) **TODO: replace/check all numbers in this caption/table for Cu+Au.**

4.4 π^0 Extraction

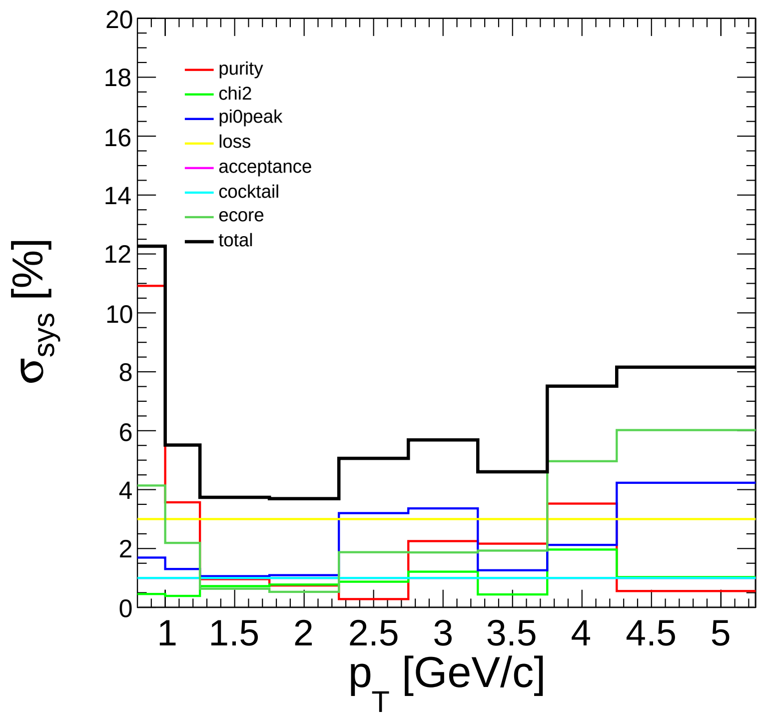
The procedure of obtaining N_{tag} consists of reconstructing π^0 signal from $M_{e^+e^-\gamma}$ distribution. After subtracting backgrounds we fit remaining distribution with Gaussian function with 2nd order polynomial. To account for uncertainty of this procedure we vary upper bound of fitting range from 0.25 to 0.35 GeV (with 0.05 step), polynomial order from 0 to 2, and upper bound of mass counting window from 0.18 to 0.21 GeV with 0.01 step (see fig. 4.1.1).

4.5 Cocktail Ratio

The second largest contributor to decay photon spectrum is η-meson, therefore the uncertainty on η/π^0 ratio is dominant in R_γ. We calculate it by varying relative yield of η at 5 GeV by 0.04 up and down. The resulting uncertainty on cocktail ratio is represented as filled area around central value (see fig. 3.6.1). We see $\sim 1\%$ effect of this variation and use this value as systematic uncertainty.

4.6 Summary of Systematic Uncertainties

Additionally to aforementioned sources we also quote uncertainty due to conversion loss of 3% and acceptance mismatch of 1% from [5].

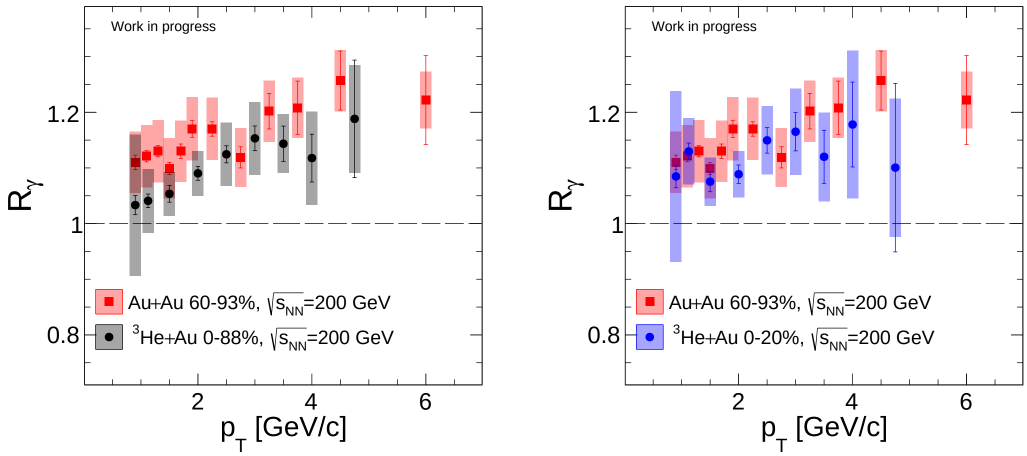


TODO (Tongzhou): replace/check this figure for the Cu+Au direct-photon analysis.

Figure 4.6.1: Summary of systematics on R_γ **TODO: replace/check all numbers in this caption/table for Cu+Au.**

5 Results

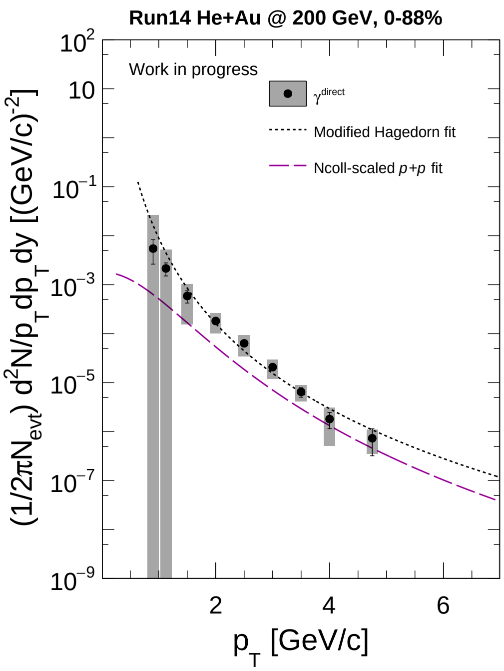
With all ingredients we can calculate R_γ (see fig. 5.0.1) and direct photon spectrum (fig. 5.0.2). Integrating invariant spectrum in range 1.0 to 5.0 GeV we get integrated yield, which is plotted on fig. 5.0.3 (pink plus markers).



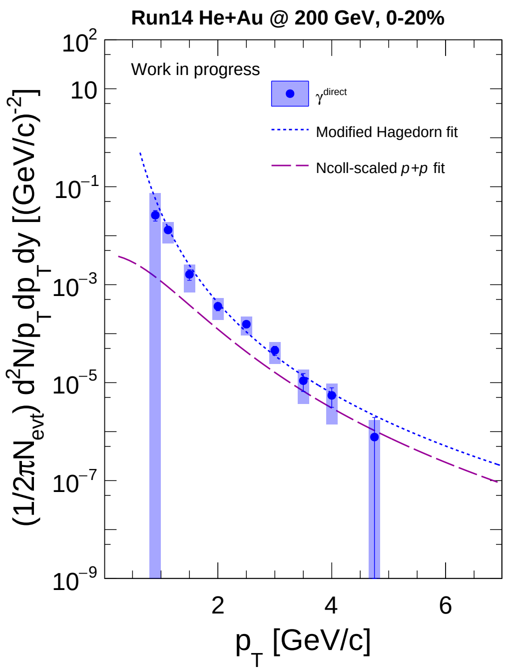
TODO (Tongzhou): replace/check this figure for the Cu+Au direct-photon analysis.

TODO (Tongzhou): replace/check this figure for the Cu+Au direct-photon analysis.

Figure 5.0.1: Measured R_γ compared with 2014 Au+Au result **TODO: replace/check all numbers in this caption/table for Cu+Au.**

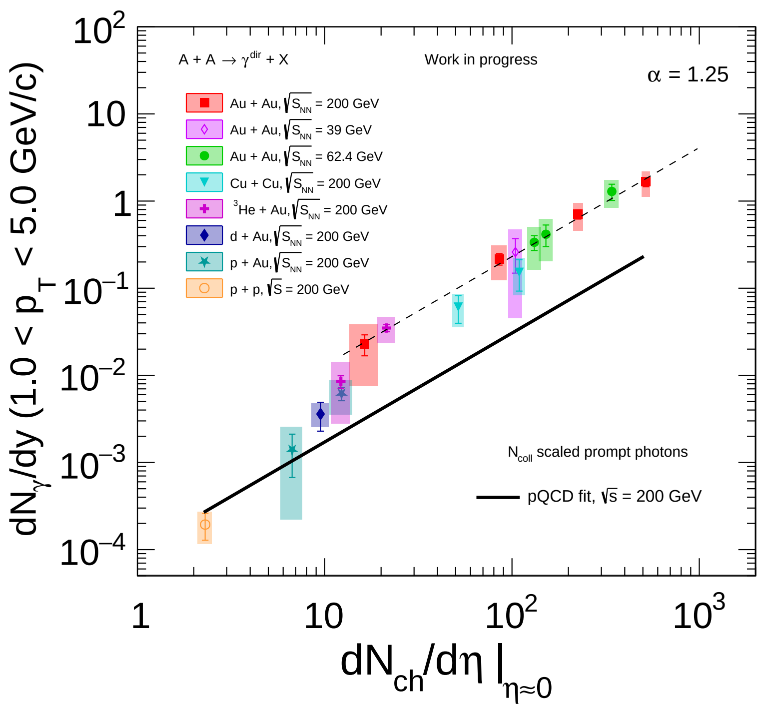


TODO (Tongzhou): replace/check this figure for the Cu+Au direct-photon analysis.



TODO (Tongzhou): replace/check this figure for the Cu+Au direct-photon analysis.

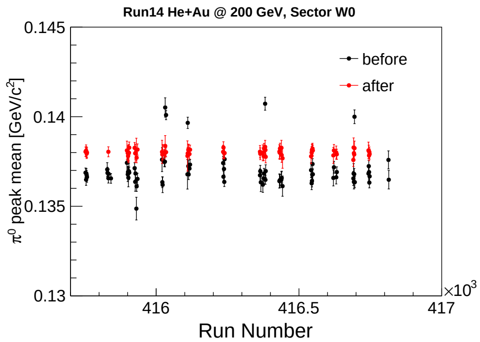
Figure 5.0.2: Invariant yield of direct photons **TODO: replace/check all numbers in this caption/table for Cu+Au.**



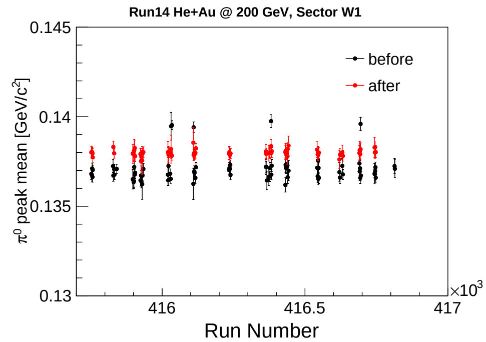
TODO (Tongzhou): replace/check this figure for the Cu+Au direct-photon analysis.

Figure 5.0.3: Integrated yield of direct photons for different collision systems as a function of charged particle multiplicity at midrapidity **TODO: replace/check all numbers in this caption/table for Cu+Au.**

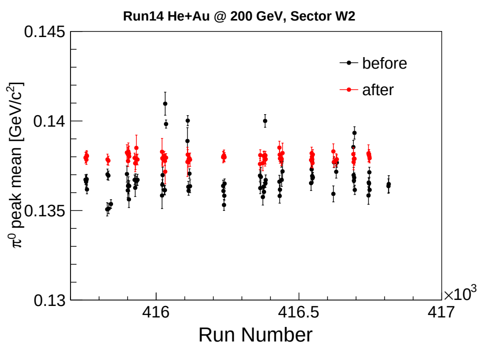
A EMCal Energy Scale Correction



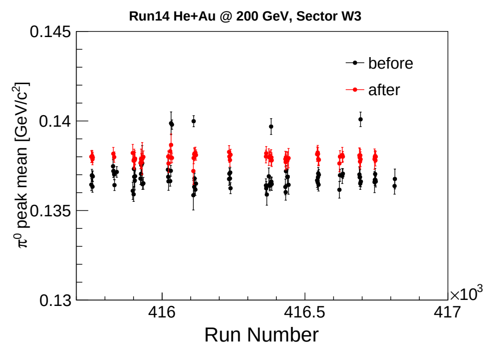
TODO (Tongzhou): replace/check this figure for the Cu+Au direct-photon analysis.



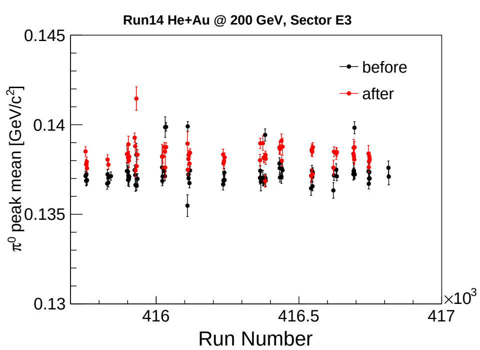
TODO (Tongzhou): replace/check this figure for the Cu+Au direct-photon analysis.



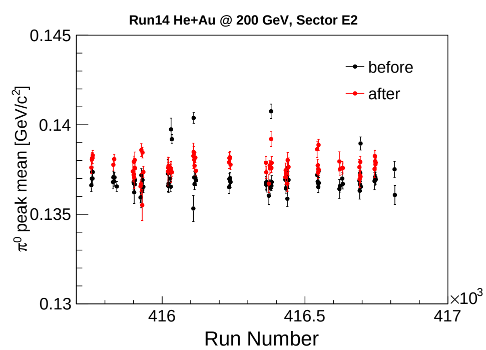
TODO (Tongzhou): replace/check this figure for the Cu+Au direct-photon analysis.



TODO (Tongzhou): replace/check this figure for the Cu+Au direct-photon analysis.



TODO (Tongzhou): replace/check this figure for the Cu+Au direct-photon analysis.



TODO (Tongzhou): replace/check this figure for the Cu+Au direct-photon analysis.

Figure A.0.1: π^0 mass dependence with respect to run number before (black points) and after (red points) energy scale calibration **TODO: replace/check all numbers in this caption/table for Cu+Au.**

B Reduced EMCal variables

Reduced EMCal variables are defined as follows

$$emcsdz(p_T) = \frac{emcdz - \langle emcdz(p_T)\rangle}{\sigma_{dz}} \tag{B.0.1}$$

$$emcsd\varphi(p_T) = \frac{emcd\varphi - \langle emcd\varphi(p_T)\rangle}{\sigma_{d\varphi}} \tag{B.0.2}$$

We fit each bin in p_T of $emcdz/emcd\varphi$ with Gaussian and then fit p_T dependencies with the following functions (separately for each EMCal sector and charge):

$$\langle emcdz(d\varphi)\rangle(p_T) = p_0 + p_1/p_T + p_2/p_T^2 + p_3/p_T^3 + p_4/p_T^4 \tag{B.0.3}$$

$$\sigma_{dz(d\varphi)} = p'_0 + p'_1/p_T + p'_2/p_T^2 + p'_3/p_T^3 \tag{B.0.4}$$

C π^0 Spectrum Fitting

We fit π^0 spectrum [2, 1] with modified Hagedorn function of the form

$$\frac{1}{2\pi p_T}\frac{d^2N}{dp_T dy} = c\left[\exp(-ap_T - bp_T^2) + p_T/p_0\right]^{-n} \tag{C.0.1}$$

Extracted parameters are listed in tab. C.1

Centrality	c	a	b	p_0	n
0-88%	133.7	0.1807	0.1216	0.6680	8.256
0-20%	249.2	0.1361	0.1476	0.6763	8.274

Table C.1: Extracted fit parameters **TODO: replace/check all numbers in this caption/table for Cu+Au.**

TODO: Check/replace the numeric values in the table above for Cu+Au.

D Energy Scale and Resolution in Simulation (EMCal)

To tune energy scale and resolution we use peak of π^0 from $\gamma\gamma$ decay channel. We require energy asymmetry of the pair to be $\alpha < 0.2$ and plot mean of the peak and its width as a function of energy. We then tune simulation constants so that both mean and width coincide with real data for $E > 2$ GeV (see tab. D.1).

Sector	W0	W1	W2	W3	E3	E2
Scale	0.994	0.986	0.990	0.988	0.989	0.986
Smear	0.063	0.088	0.077	0.071	0.063	0.068

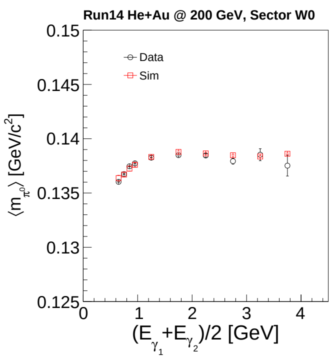
Table D.1: Scaling constants for simulation **TODO: replace/check all numbers in this caption/table for Cu+Au.**

TODO: Check/replace the numeric values in the table above for Cu+Au.

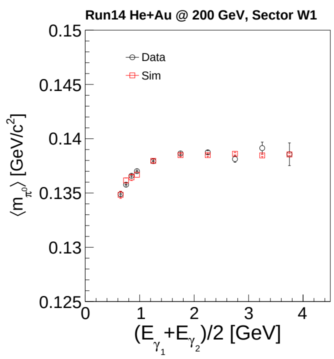
Next we fit ratio of $m_{\pi^0}^{\mathrm{sim}}/m_{\pi^0}^{\mathrm{real}}$ (all PbSc sectors combined) with functional from

$$f(E) = \frac{a}{1 + b\exp(-cE)} \tag{D.0.1}$$

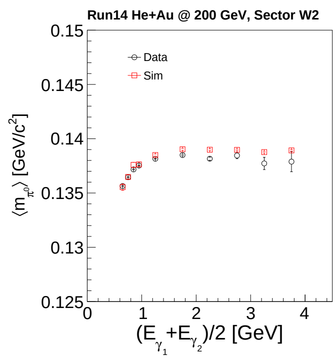
to capture lower energy region better. Extracted parameters: $a = 1.000$, $b = 0.08978$, $c = 2.4476$.



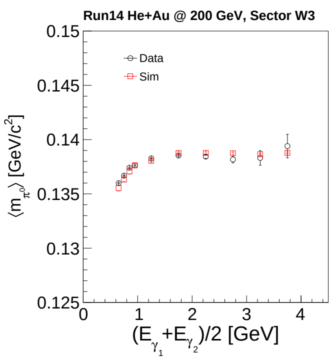
TODO (Tongzhou): replace/check this figure for the Cu+Au direct-photon analysis.



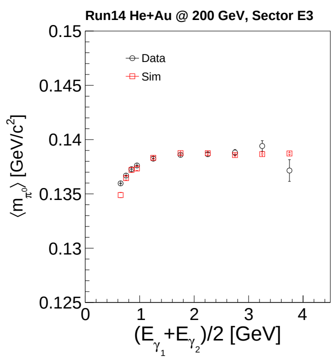
TODO (Tongzhou): replace/check this figure for the Cu+Au direct-photon analysis.



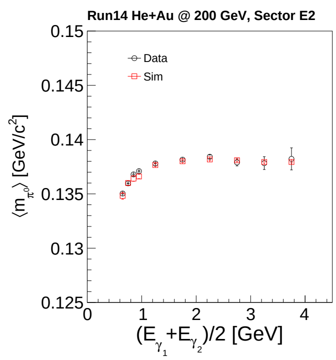
TODO (Tongzhou): replace/check this figure for the Cu+Au direct-photon analysis.



TODO (Tongzhou): replace/check this figure for the Cu+Au direct-photon analysis.

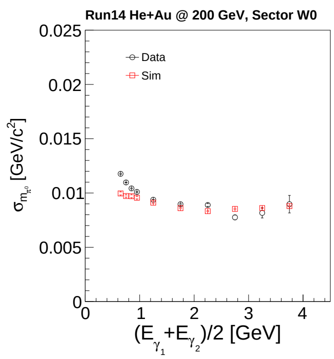


TODO (Tongzhou): replace/check this figure for the Cu+Au direct-photon analysis.

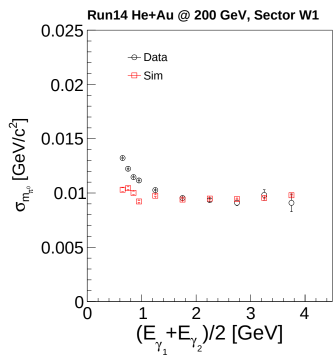


TODO (Tongzhou): replace/check this figure for the Cu+Au direct-photon analysis.

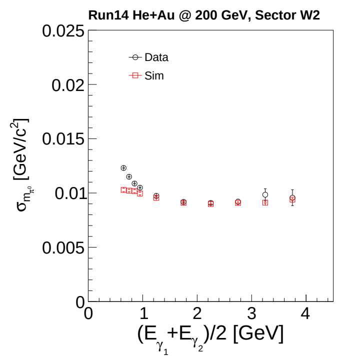
Figure D.0.1: Mean of π^0 peak as a function of energy for simulation and real data **TODO: replace/check all numbers in this caption/table for Cu+Au.**



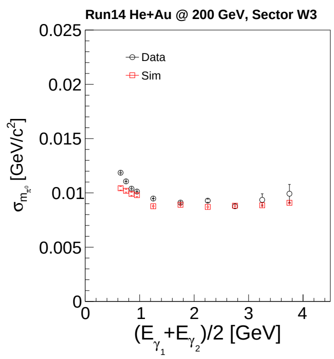
TODO (Tongzhou): replace/check this figure for the Cu+Au direct-photon analysis.



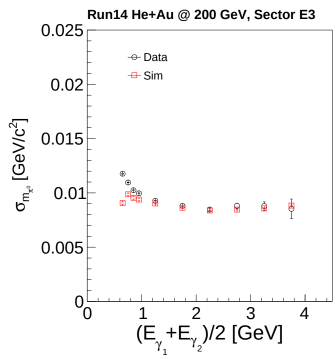
TODO (Tongzhou): replace/check this figure for the Cu+Au direct-photon analysis.



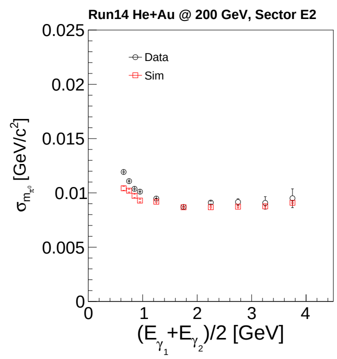
TODO (Tongzhou): replace/check this figure for the Cu+Au direct-photon analysis.



TODO (Tongzhou): replace/check this figure for the Cu+Au direct-photon analysis.



TODO (Tongzhou): replace/check this figure for the Cu+Au direct-photon analysis.



TODO (Tongzhou): replace/check this figure for the Cu+Au direct-photon analysis.

Figure D.0.2: Width of π^0 peak as a function of energy for simulation and real data **TODO: replace/check all numbers in this caption/table for Cu+Au.**

E Data Tables

p_T, GeV/c	R_γ	Stat.	Sys.
0.900	1.0332	0.0172	0.1267
1.125	1.0409	0.0122	0.0574
1.500	1.0535	0.0151	0.0394
2.000	1.0903	0.0125	0.0403
2.500	1.1245	0.0153	0.0569
3.000	1.1532	0.0222	0.0656
3.500	1.1435	0.0317	0.0526
4.000	1.1177	0.0430	0.0840
4.750	1.1882	0.1056	0.0969

Table E.1: Excess ratio R_γ in Cu + Au collision system at $\sqrt{s_{NN}} = 200$ GeV for 0-88% centrality class **TODO: replace/check all numbers in this caption/table for Cu+Au.**

TODO: Check/replace the numeric values in the table above for Cu+Au.

p_T, GeV/c	R_γ	Stat.	Sys.
0.900	1.0847	0.0204	0.1535
1.125	1.1296	0.0150	0.0597
1.500	1.0754	0.0182	0.0432
2.000	1.0887	0.0165	0.0415
2.500	1.1495	0.0230	0.0615
3.000	1.1650	0.0344	0.0776
3.500	1.1200	0.0477	0.0799
4.000	1.1780	0.0763	0.1326
4.750	1.1005	0.1514	0.1243

Table E.2: Excess ratio R_γ in Cu + Au collision system at $\sqrt{s_{NN}} = 200$ GeV for 0-20% centrality class **TODO: replace/check all numbers in this caption/table for Cu+Au.**

TODO: Check/replace the numeric values in the table above for Cu+Au.
TODO: Check/replace the numeric values in the table above for Cu+Au.
TODO: Check/replace the numeric values in the table above for Cu+Au.

p_T, GeV/c	Inv. Yield	Stat.	Sys.
0.900	5.4644e-03	2.8373e-03	2.0843e-02
1.125	2.1455e-03	6.3816e-04	3.0103e-03
1.500	5.8703e-04	1.6520e-04	4.3208e-04
2.000	1.8155e-04	2.5167e-05	8.0963e-05
2.500	6.3501e-05	7.8306e-06	2.9034e-05
3.000	2.0757e-05	3.0124e-06	8.8867e-06
3.500	6.4692e-06	1.4312e-06	2.3741e-06
4.000	1.8090e-06	6.6094e-07	1.2908e-06
4.750	7.3318e-07	4.1147e-07	3.7762e-07

Table E.3: Direct photon invariant yield in Cu + Au collision system at $\sqrt{s_{NN}} = 200$ GeV for 0-88% centrality class **TODO: replace/check all numbers in this caption/table for Cu+Au.**

p_T, GeV/c	Inv. Yield	Stat.	Sys.
0.900	2.6224e-02	6.3221e-03	4.7493e-02
1.125	1.3097e-02	1.5185e-03	6.0311e-03
1.500	1.6216e-03	3.9173e-04	9.2890e-04
2.000	3.6143e-04	6.7414e-05	1.6908e-04
2.500	1.5613e-04	2.3978e-05	6.4224e-05
3.000	4.5618e-05	9.5090e-06	2.1454e-05
3.500	1.0970e-05	4.3565e-06	7.3072e-06
4.000	5.4931e-06	2.3541e-06	4.0923e-06
4.750	7.7542e-07	1.1678e-06	9.5842e-07

Table E.4: Direct photon invariant yield in Cu + Au collision system at $\sqrt{s_{NN}} = 200$ GeV for 0-20% centrality class **TODO: replace/check all numbers in this caption/table for Cu+Au.**

References

[1] U. A. Acharya et al. “Systematic study of nuclear effects in p +Al, p +Au, d +Au, and ^{3}Cu+Au collisions at $\sqrt{s_{_{NN}}}$ = 200 GeV using π^0 production”. In: *HEPData* (2021). DOI: 10.17182/hepdata.115023.

[2] U.A. Acharya et al. “Systematic study of nuclear effects in p +Al, p +Au, d +Au, and ^{3}Cu+Au collisions at $\sqrt{s_{_{NN}}}$ = 200 GeV using π^0 production”. In: *Phys. Rev. C* 105.6 (2022), p. 064902. DOI: 10.1103/PhysRevC.105.064902. arXiv: 2111.05756 [nucl-ex].

[3] *Analysis Note 1270, Measurement of* π^0 *in* $\sqrt{s_{NN}} = 200$ *GeV Cu+Au collisions.*

[4] *Analysis Note 1399,* $\phi \rightarrow K^+K^-$ *measurements in Run14 Cu+Au, Run15 p+Au and p+Al at* $\sqrt{s_{NN}} = 200$ *GeV.*

[5] *Analysis Note 1417, Low-*p_T *Direct Photon Production in Au+Au Collisions at 200 GeV Beam Energy.*

[6] *Analysis Note 1448, Final Results on Low-*p_T *Direct Photon Production in Au+Au Collisions at 200 GeV Beam Energy.*

[7] *Analysis Note 1501, Measurements of* η*-meson production in Run14* $Cu + Au$ *at* $\sqrt{s_{_{NN}}} = 200$ *GeV.*

[8] *Technical Note 481, Track Reconstruction of External Conversions.*